PRABHAT SINGH

Email | Phone | LinkedIn | GitHub | X

SKILLS

- **Languages & Backend:** Python, asyncio, FastAPI, Websockets, PostgreSQL, SQLAlchemy, Pydantic
- **Agentic AI & Orchestration:** Agent Loop Design, Multi-Agent Systems, Tool/Function Calling, Shared-State Orchestration, MCP, Context Management, Short-Term & Long-Term Memory, Mem0
- **RAG & Retrieval:** RAG Pipelines, Query Routing & Decomposition, Embeddings, Vector Search, Context Grounding, Citation-Based Answers, LLM Output Verification
- **LLMs & Vector Databases:** Gemini, Qwen, GPT-OSS, ChromaDB
- **Security & Systems :** AES-256 encryption, RBAC, Active Directory/LDAP, Nextcloud API, file system monitoring
- **Tools & Platforms :** Git, GitHub, Docker, Linux, Claude Code, Postman, pytest

PROJECTS

Multi-Agent RAG System | Python, asyncio, Gemini, ChromaDB AgenticRAG-github

- Built a multi-agent RAG system over PDFs in plain asyncio, no framework: Router, Retriever, Synthesizer and Critic agents coordinate only through a shared state, sequenced by one orchestrator.
- Router classifies queries as no-, single- or multi-retrieval and generates sub-queries; Retriever merges and dedupes ChromaDB results using asymmetric Gemini embeddings and per-page chunking for exact file and page citations.
- Critic verifies each answer's claims against retrieved chunks (Supported / Partially / Unsupported); unsupported answers trigger one bounded retry with a follow-up query, cutting hallucinations without infinite loops.
- Added per-agent tracing and structured JSON outputs with safe fallbacks, so every run shows which agents ran, what was retrieved, and the verdict.

Coding Agent Harness | Python, OpenRouter, Mem0, MCP Engram-github

- Built Engram, a lightweight terminal-based coding agent that streams LLM responses and executes tools for reading, writing, editing, and running shell commands through a custom agent loop, without external agent-framework dependencies.
- Used OpenRouter as a provider-agnostic model layer, enabling model switching without code changes.
- Integrated GitHub through MCP, allowing the agent to work with existing codebases: browsing repositories, reading and editing files, and managing issues and pull requests.
- Integrated Mem0 for persistent memory, enabling the agent to retain project context and user preferences across sessions.

EXPERIENCE

Software Developer

WESEE (Research and development lab) | May 2024 - August 2025

- Built CloudGuard, FastAPI based web application integrated with Nextcloud APIs for secure file uploads, downloads, synchronization and sharing across local and cloud storage.
- Implemented AES-256 file encryption with custom key management and integrated Active Directory/LDAP authentication with role based access control(RBAC), ensuring that only encrypted files were stored in Nextcloud and decryption was restricted to authorized users.
- Developed a watchdog-based folder monitor that automatically encrypts newly added files, along with double-click-triggered decryption and timed deletion of decrypted files.

EDUCATION

- **Super30 Program | 100xSchool | May 2026 – Present**
  Selective, cohort-based program focused on building AI/Agentic systems
- **M.Sc in Computer Science (AI and Machine Learning)**
  Scaler Neovarsity | Expected Dec 2026
- **M.Tech in Aerospace (Flight Mechanics and Control)**
  IIT Kanpur | 2022
- **B.Tech in Electrical and Electronics Engineering**
  Galgotias College of Engineering and Technology | 2018